

Measurement of Human Brown Adipose Tissue Volume and Activity Using Anatomical MRI and Functional MRI

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The existence of brown adipose tissue (BAT) in humans has previously been assessed in vivo via sequential ^{18}F -FDG PET/CT imaging. We developed a MRI protocol to detect BAT mass based on BAT's property of having higher water-to-fat ratio than white adipose tissue (WAT). We showed that the signal contrast obtained between water-saturation and without water-saturation was higher in BAT than in WAT in fast spin echo images and in T2-weighted images. The water-to-fat ratio was also higher in BAT via contrasting the water and fat images of the Dixon method. The MRI measured volume and location of BAT was similar to PET/CT results in the same subjects. In addition, we also demonstrated that cold challenges (14 °C) led to significant fMRI BOLD signal increases in BAT.